

The compact $\text{Spin}(7)$ sigma-MGF on S_8

Proof and exact degree-four matrix verification

Computational proof companion

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Abstract

For the complexification of the real eight-dimensional spin representation, we prove the compact Molien–Weyl formula, construct explicit maximal-torus matrices, and verify all coefficients through total degree four by exact constant terms. In the Schur basis the answer is $1 + h_2 + h_4 + s_{(2,2)} + e_4 + O_{\geq 6}$. The coefficient of $m_{(1,1,1,1)}$ is therefore 4, with the extra invariant supplied by the Cayley alternating four-form.

1 The coefficient formula

Let K be compact, let V be a finite-dimensional unitary K -module, and write $h_d(\rho(g))$ for the complete homogeneous character of degree d . Expanding one determinant at a time gives the formal identity

$$\prod_{a \geq 1} \det(I - t_a \rho(g))^{-1} = \prod_{a \geq 1} \sum_{d \geq 0} h_d(\rho(g)) t_a^d.$$

The coefficient of the monomial symmetric function m_τ , for $\tau = (\tau_1, \dots, \tau_r)$, is consequently

$$\int_K \prod_{j=1}^r h_{\tau_j}(\rho(g)) dg = \dim \left(\bigotimes_{j=1}^r \text{Sym}^{\tau_j} V \right)^K.$$

This proves the proposed formula

$$\mathcal{S}_{K,V} = \int_K \prod_a \det(I - t_a \rho(g))^{-1} dg = \sum_{\tau} \dim \left(\bigotimes_j \text{Sym}^{\tau_j} V \right)^K m_{\tau}.$$

Equivalently, the Cauchy identity gives the Schur-basis expansion

$$\mathcal{S}_{K,V} = \sum_{\lambda} \dim(\mathbb{S}_{\lambda} V)^K s_{\lambda}.$$

The notebook checks both bases independently.

2 Matrix model and Weyl constant term

Use type B_3 with orthonormal coordinates e_1, e_2, e_3 . The weights of S_8 are

$$\text{Wt}(S_8) = \left\{ \frac{1}{2}(\epsilon_1 e_1 + \epsilon_2 e_2 + \epsilon_3 e_3) : \epsilon_i \in \{\pm 1\} \right\}.$$

For $\theta \in \mathbb{R}^3$ this constructs the explicit unitary matrix

$$M(\theta) = \text{diag}_{\epsilon \in \{\pm 1\}^3} \exp \left(\frac{i}{2} \sum_{j=1}^3 \epsilon_j \theta_j \right) \in U(8).$$

These matrices form the image of a maximal torus of Spin(7). Together with the 18 roots and $|W(B_3)| = 48$, the Weyl integration formula turns the group integral into the exact constant term

$$\mathcal{S}_{\text{Spin}(7),8} = \frac{1}{48} \text{CT}_z \left[\prod_{\alpha > 0} (1 - z^\alpha)(1 - z^{-\alpha}) \prod_{a,\mu \in \text{Wt}(S_8)} (1 - t_a z^\mu)^{-1} \right].$$

Thus it is unnecessary to approximate Haar measure by random matrices.

3 Predicted low-degree tensors

The central element $-1 \in \text{Spin}(7)$ acts by $-I$ on S_8 , so all odd-total-degree coefficients vanish. The tensor square is

$$S_8 \otimes S_8 = \mathbf{1} \oplus \mathbf{7} \oplus \mathbf{21} \oplus \mathbf{35}, \quad \text{Sym}^2 S_8 = \mathbf{1} \oplus \mathbf{35}, \quad \bigwedge^2 S_8 = \mathbf{7} \oplus \mathbf{21}.$$

It is multiplicity-free, hence

$$\dim(S_8^{\otimes 4})^{\text{Spin}(7)} = \dim \text{End}_{\text{Spin}(7)}(S_8 \otimes S_8) = 4.$$

Three tensors are the metric pairings. The fourth is the Cayley form in $\bigwedge^4 S_8^*$. In Schur notation this yields

$$\mathcal{S}_{\text{Spin}(7),8} = 1 + h_2 + h_4 + s_{(2,2)} + e_4 + O_{\geq 6}.$$

Changing to the monomial basis gives, through degree four,

$$\frac{\tau}{[m_\tau]} \left| \begin{array}{cccccccc} (2) & (1,1) & (4) & (3,1) & (2,2) & (2,1,1) & (1,1,1,1) \\ 1 & 1 & 1 & 1 & 2 & 2 & 4 \end{array} \right|$$

Every coefficient of total degree one or three is zero.

4 Independent exact verification

The marimo notebook generates the B_3 root system and the eight-weight orbit from the Cartan matrix. It then performs two distinct checks. First, for a concrete $M(\theta)$ it compares

$$\prod_a \det(I - t_a M(\theta))^{-1} \quad \text{with} \quad \prod_{a,\mu} (1 - t_a e^{i\langle \mu, \theta \rangle})^{-1}.$$

Second, it forms the characters $h_{\tau_1} \cdots h_{\tau_r}$ directly from the weights and applies the one-sided Weyl-denominator identity

$$\prod_{\alpha > 0} (1 - z^{-\alpha}) = \sum_{w \in W} \det(w) z^{w\rho - \rho}.$$

The resulting constant terms are exact integers and do not use the quoted tensor-square decomposition as input. The exact output is

τ	(2)	(1,1)	(4)	(3,1)	(2,2)	(2,1,1)	(1,1,1,1)
proposed	1	1	1	1	2	2	4
Weyl check	1	1	1	1	2	2	4

with every omitted partition of degree at most four returning zero. In the Schur basis the nonzero multiplicities are

$$[s_{(2)}] = [s_{(4)}] = [s_{(2,2)}] = [s_{(1,1,1,1)}] = 1.$$

Reproducibility. Run the marimo notebook `spin7_verification.py`. Its shared exact-arithmetic core is in the parent folder. The test suite also executes this degree-four case noninteractively.